Graves' Disease (GD)

Karen Pendergrass, Giorgos Aristotelous

Microbiome Signatures, Microbiome Medicine

Abstract. OverviewGraves' Disease (GD) affects approximately 0.5% of the population, predominantly women. First-line treatment options—antithyroid medications, radioactive iodine, and surgery— often result in significant side effects, incomplete remissions, and frequent relapses. Further, current first-line treatment options focus on symptoms management, and reflect an inadequate understanding of the etiology of the condition. However, recent research reveals a [...]

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Overview

Graves' Disease (GD) affects approximately 0.5% of the population, predominantly women. First-line treatment options—antithyroid medications, radioactive iodine, and surgery— often result in significant side effects, incomplete remissions, and frequent relapses. Further, current first-line treatment options focus on symptoms management, and reflect an inadequate understanding of the etiology of the condition. However, recent research reveals a strong causal relationship between GD and the gut microbiome.[1] Thus, a microbiome-targeted strategy aimed at addressing the root causes of GD moves beyond mere symptom suppression and marks a significant advancement in medical history.

Associated Conditions

Graves' disease is a complex autoimmune condition primarily affecting the thyroid gland, leading to hyperthyroidism, but it is also associated with a wide range of other conditions and symptoms, such as Inflammatory Bowel Disease (IBD), Crohn's disease (CD), depression, and Rheumatoid Arthritis (RA). [2][3][4] These associated conditions can also influence the development or progression of GD. The shared microbiome alterations between Graves' Disease and these other conditions underscore the interconnectedness of systemic health, particularly the interplay between the gut, immune system, and endocrine function. Understanding these overlaps opens new avenues for microbiome-targeted interventions (MBTIs) as potential therapeutic or preventative strategies for GD and its comorbidities.

▶ **What other conditions are associated with Grave's Disease?**

Causes

Graves' disease is an organ-specific autoimmune thyroid disorder characterized by hyperthyroidism driven by autoantibodies against the thyroid-stimulating hormone receptor (TSHR). Classic paradigms emphasize a genetic predisposition interacting with environmental triggers to precipitate a loss of immune tolerance.

Indeed, genetic susceptibility, combined with factors like iodine excess, psychosocial stress, infections, and postpartum immune shifts, are known to contribute to GD development in susceptible individuals.[16][17] However, these factors alone do not fully explain disease onset, as many exposed individuals never develop GD and the precise pathogenic mechanism is unclear. Emerging research has expanded the scope of inquiry to the "thyroid–gut axis," investigating how gut microbiome dysbiosis and increased intestinal permeability ("leaky gut") might instigate or amplify autoimmune processes. Notably, recent Mendelian randomization studies provide evidence supporting a causal link between gut microbial profiles and GD risk, underscoring the complex interplay between genes, immunity, and the environment.[18] The table below summarizes prevailing causal theories of GD and highlights key limitations or criticisms for each, based on current (post-2020) evidence.

▶ **What are the current causal theories of Grave's Disease and their limitations?**

Diagnosis

Primer

Understanding metal homeostasis and mineral homeostasis is essential for fully elucidating Graves' Disease, and helps us grasp the broader implications of the condition's unique microbiome signature.

Metal Homeostasis

Data reveals that elevated levels of cadmium (Cd), lead (Pb), and chromium (Cr) are associated with an increased risk of hyperthyroidism, as these metals can disrupt endocrine functions and cause oxidative damage to the thyroid gland. These toxic elements pose significant health risks due to their cumulative nature. Conversely, higher cobalt (Co) levels are associated with a decreased risk of hyperthyroidism, highlighting its complex role in thyroid metabolism. Additionally, deficiencies in copper (Cu) and zinc (Zn), essential for thyroid hormone synthesis, are also associated with hyperthyroidism and GD. [19] While these findings certainly aid in understanding occupational risk factors for GD, further research provides alarming new insight on metal toxicity and carcinogenicity occurring in thyroid cells when chronically exposed to metal concentrations that are slightly increased, even within what is considered the "normal" range. [20]

▶ **What occupations are risk factors associated with Grave's Disease due to exposure?**

Mineral Homeostasis

Extensive research shows the thyroid-gut axis (TGA) significantly influences thyroid function. Gut health regulates thyroid roles and pathologies through nutrient intake and microbiota. Essential minerals like iodine and selenium are critical for thyroid hormone synthesis and overall health. The gut-immune interaction affects autoimmune diseases, including Graves' disease and thyroid cancer.

▶ **What minerals are involved in Grave's Disease and what are their functions?**

Interventions

Researchers involved in the microbiota analyses of GD predict that microbiota-targeted therapeutics will emerge as the new strategy for managing GD/GO in the coming years. [24] The section covers interventions for Grave's Disease, such as pharmacological treatments, drug repurposing, and dietary supplements. It explores ASAPs, utilizing emerging science to find new treatments, and STOPs, which suggest reassessing standard practices. Here we suggest other microbiome-targeted interventions (MBTIs) for the management of GD.

Pharmacological

The therapeutic approach to Graves' disease (GD) comprises thionamides, radioiodine ablation, or surgery as first-line therapy, and cholestyramine and oral iodine as second-line therapies.

First-Line Pharmacological Treatments for Grave's Disease

Methimazole: Methimazole up-regulates the levels of *Bifidobacterium* and *Collinsella* which are decreased in GD, and down-regulates the levels of *Prevotella* and *Dialister*, which are increased in GD. [x]

Serum from patients with untreated Graves' disease had a significantly higher concentration of Cu, Zn-SOD and higher SOD-like activity than those from normal subjects. This is likely due to the presence of *Haemophilus parainfluenzae*, which is significantly increased in Grave's Disease, that has a rare Copper-zinc superoxide dismutase ([Cu, Zn]-SOD) encoding. [25][x]

It is noteworthy that methimazole treatment produced no significant change in SOD-like activity and Cu, Zn-SOD concentration in patients with Graves' disease.[26] Thus, it is likely that a patient not responding to Methimazole treatment has an increased level of *H. parainfluenzae*.

▶ **What are the risks associated with first-line therapies for GD?**

Nonthionamide antithyroid drugs (NTADs)

Although thionamide antithyroid drugs are the cornerstone of hyperthyroidism treatment, some patients cannot tolerate this drug class because of its serious side effects including agranulocytosis, hepatotoxicity, and vasculitis. Therefore, non-thionamide antithyroid drugs (NTADs) play an important role in controlling hyperthyroidism in clinical practice.

▶ **What NTADs have been investigated for Grave's Disease and hyperthyroidism?**

Drug Repurposing

Metformin: Metformin has shown promising benefits for treating Graves' ophthalmopathy (GO) by activating the AMPK/mTOR pathway, which reduces inflammation and fibrosis in orbital tissues. [31] This suggests metformin could improve symptoms and reduce reliance on antithyroid drugs and other treatments with serious potential side effects. Additionally, metformin increases the abundance of Akkermansia muciniphila— [32] a beneficial bacterium that is decreased in GD— and reduced orbital pathology in GO patients has been positively correlated with higher levels of Akkermansia. [33]

Rituximab: Research suggests that roughly 70% of GD patients have evidence of Graves' orbitopathy (GO). [34]

Antibiotics

The oral administration of the antibiotic vancomycin reduced the severity of GD/GO in mouse models. [35]

Supplements

Vitamin D: Research has increasingly highlighted the significant role of vitamin D in the pathogenesis and management of Graves' Disease (GD). Numerous studies have observed that patients with GD often exhibit lower levels of vitamin D compared to healthy individuals. Vitamin D is known to modulate the activity of immune cells, such as T and B lymphocytes, dendritic cells, and monocytes, which are crucial in the autoimmune response seen in GD. By regulating these cells, vitamin D can help reduce the production of proinflammatory cytokines and enhance anti-inflammatory cytokines, thereby possibly mitigating the autoimmune attack on the thyroid. [36][x] Clinical trials and meta-analyses also suggest that vitamin D supplementation reduces the relapse rate of GD after antithyroid drug (ATD) treatment. [37]

Inositol: Research findings indicate that a significant proportion of Anaerostipes species, which are decreased in the Mircobiome Signature of Grave's Disease, can convert inositol into propionate. [38] Another study highlighted that Myo-inositol and selenium (Myo-Ins-Se) supplementation effectively normalized thyroid-stimulating hormone (TSH) levels and improved overall thyroid function in hyperthyroid and hypothyroid patients. This combined therapy showed promise in restoring euthyroidism, potentially offering a new, effective treatment for Graves' Disease and other forms of hyperthyroidism and hypothyroidism. [39]

Selenium: Selenium deficiency has been linked to autoimmune thyroid diseases, including GD. Adequate selenium levels help mitigate the oxidative stress associated with thyroid hormone production and can support overall thyroid health. Supplementation of selenium has been shown to improve thyroid function and reduce symptoms in patients with GD. [40] For patients with Graves' disease, especially those with mild thyroid eye disease, selenium supplementation may lead to faster remission of hyperthyroidism, improved quality of life, and better eye involvement outcomes. [x]

Glutathione: Exposure to lead, a redox inactive metal, depletes the cell's major antioxidant reserves of glutathione.

Berberine: Methimazole combined with berberine achieves a better effect on GD than methimazole alone, and the changes in the gut microbiome include changes in levels of *Lactococcus lactis*, *Enterobacter hormaechei*, and *Chryseobacterium indologenes*. [42]

B-12: Vitamin B-12 (B-12) has the largest and most chemically complex structure of all of the vitamins. Also called cobalamin, B-12 is the only active substance in the body containing an atom of cobalt. [43] As mentioned previously, higher cobalt (Co) levels are associated with a decreased risk of hyperthyroidism, and may have protective effects.

Lifestyle Interventions

A structured exercise program in euthyroid Graves' disease patients improved aerobic capacity, reduced fatigue, normalized some thyroid hormones, accelerated anti-thyroid medication withdrawal, and reduced relapse rates, highlighting exercise's clinical and potential immunological benefits. [44]

STOPs

While probiotics often hold potential in managing GD/GO, careful selection and optimization of microbial strains are crucial. Probiotic formulas should exclude strains such as Lactobacillus, Prevotella, and Veillonella, which are often found in increased abundance in GD patients.

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Microbiome Signature: Graves' Disease (GD)

Research Feed

Assessment of Thyroid Function and Oxidative Stress State in Foundry Workers Exposed to Lead
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Americans are over three times more likely to suffer from autoimmune diseases compared to the global average, with approximately 16.67% of the U.S. population affected versus 5% worldwide.

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Graves' Disease (GD) •
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Did you know?
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Irritable Bowel Syndrome (IBS) •
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Irritable Bowel Syndrome (IBS) is a common gastrointestinal disorder characterized by symptoms such as abdominal pain, bloating, and altered bowel habits. Recent research has focused on the gut microbiota's role in IBS, aiming to identify specific microbial signatures associated with the condition.

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Metals •
Metals

Heavy metals influence microbial pathogenicity in two ways: they can be toxic to microbes by disrupting cellular functions and inducing oxidative stress, and they can be exploited by pathogens to enhance survival, resist treatment, and evade immunity. Understanding metal–microbe interactions supports better antimicrobial and public health strategies.

Microbes •

Microbes

Microbes are microscopic organisms living in and on the human body, shaping health through digestion, vitamin production, and immune protection. When microbial balance is disrupted, disease can occur. This guide explains the key types of microorganisms—bacteria, viruses, fungi, protozoa, and archaea—along with major examples of pathogenic and beneficial species.

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Trace Elements and Thyroid Diseases: Comprehensive Review of Endocrine and Microbiome Interactions

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Targeted Therapies and Microbiome Insights in Graves' Disease: A Clinical Review
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Microbiome-Targeted Interventions (MBTIs)
Microbiome Targeted Interventions (MBTIs) are cutting-edge treatments that utilize information from Microbiome Signatures to modulate the microbiome, revolutionizing medicine with unparalleled precision and impact.

Metal Homeostasis

Transition metals like iron, zinc, copper, and manganese are crucial for the enzymatic machinery of organisms, but their imbalance can foster pathogenic environments within the gastrointestinal tract.

Zinc

Zinc is an essential trace element vital for cellular functions and microbiome health. It influences immune regulation, pathogen virulence, and disease progression in conditions like IBS and breast cancer. Pathogens exploit zinc for survival, while therapeutic zinc chelation can suppress virulence, rebalance the microbiome, and offer potential treatments for inflammatory and degenerative diseases.

Cholestyramine

Cholestyramine, a polymer resin, binds bile acids, toxins, and heavy metals, reducing cholesterol and fat absorption, while altering gut microbiome and aiding detoxification.

References

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Increased Info

Decreased Info
Increased
Decreased
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Relative Abundance (Class)
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Relative Abundance (Family)
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Relative Abundance (Genus)
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Relative Abundance (Kingdom)
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Relative Abundance (Order)
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Relative Abundance (Phylum)
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Relative Abundance (Species)

▶ **Studies**

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